

Problem 9.17

A fund earns a nominal rate of interest of 6% compounded every two years. Calculate the amount that must be contributed now to have 1000 at the end of six years.

Solution

Let x be the amount contributed now. Since the nominal rate of interest is 6% compounded every two years, the two-year effective rate is

$$2i = 2(0.06) = 0.12.$$

Six years contains three two-year periods, so the accumulated value equation is

$$1000 = x(1 + 2i)^3.$$

Solving for x ,

$$x = \frac{1000}{(1 + 2i)^3}.$$

Substituting $i = 0.06$ gives

$$x = \frac{1000}{(1 + 2(0.06))^3}.$$

Thus,

$$x = \frac{1000}{(1.12)^3}.$$

Finally,

$$x \approx 711.780247.$$

Hence, the amount that must be contributed now is

$$\boxed{711.78}.$$